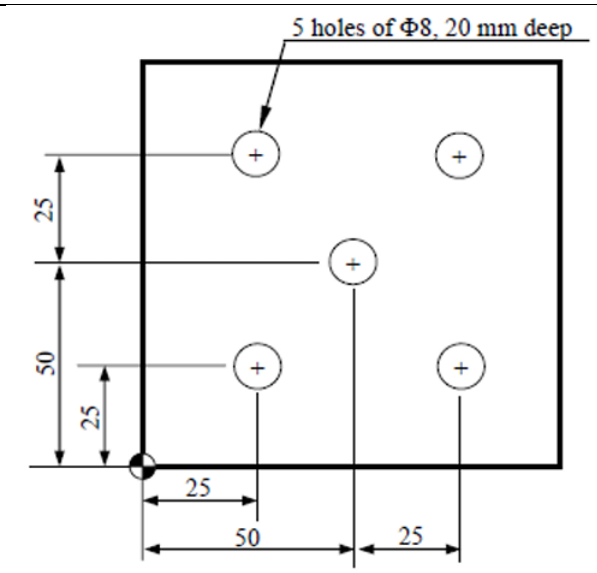
MANIPAL INSTITUTE OF TECHNOLOGY
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(A constituent unit of MAHE, Manipal)

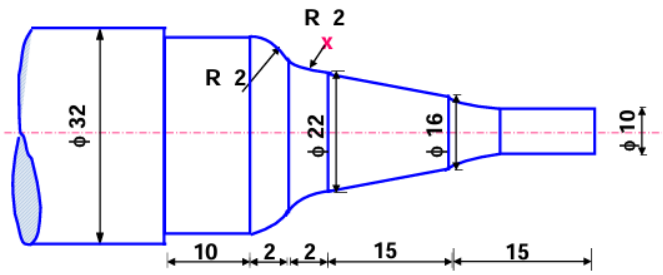
DEPARTMENT OF AERONAUTICAL ENGINEERING & AUTOMOBILE ENGINEERING			
END SEMESTER EXAMINATION		**DECEMBER 2024**	
Level	**B.Tech**	**Curriculum**	**2022**
Program Name	**AERONAUTICAL ENGINEERING**	**Course Type**	**C/FC/PE/OE**
Course Name	**Materials and Processing Techniques**	**Course Code**	**AAE 2122**
Semester	**III**	**Max Marks**	**50**
Date	**02-12-2024**	**Duration**	**3 Hours**

Instructions:

- **Answer ALL the questions.**
- **Data not provided may be suitably assumed.**

Q. No.	Question	Marks	NBA		IET*		BL
			CO	PO	CLO	AHEP4 LO	
1.	Define rapid prototyping and explain its importance in the aerospace industry.	3	4	1,2,3,5	3	**1,4,5,7,13**	**2**
2.	What are the Miller indices for a plane that cuts the x-axis at 1 unit, the y-axis at 2 units, and the z-axis at 3 units?	2	1	1	2	**1,2**	**2**
3.	With relevant sketches list and describe the primary steps involved in the powder metallurgy process.	5	4	1,3,5	5	**1,3,12,13**	**2**
4.	With a relevant sketch explain the peritectic reaction in the iron-carbon phase diagram and list all the invariant reactions in the iron-carbon phase diagram.	4	3	1,2	4	**1,2,4**	**4**
5.	What is diffusion, and what are the factors that influence diffusion in solids?	3	2	1,2,4	3	**1,2,7**	**2**
6.	Write a part program for the component given below.	3	4	5,10	3	**3,13**	**3**

							
47.	Discuss the role and importance of the nozzle in Abrasive Jet Machining, including factors affecting its life span.	3	4	1,2,5,6	3	**4,7,13**	**4**
84.	What are dislocations, and how do they affect the mechanical properties of materials?	3	2	1,2	3	**1,2,7**	**4**
9.2	List the different types of solid solutions. With a neat sketch explain the ordered substitutional solid solution	4	1	1,2,3	4	**1,2,3**	**2**
10.	What are the advantages of using stereolithography (SLA) in rapid prototyping?	3	4	1,2,3,5	3	**1,5,7,13**	**2**
11.	Two metals A and B have their melting points at 650 C and 450 C respectively. These metals do not form any compound or intermediate phase. The maximum solubility of A in B and B in A is 10% at 300 C, 10% and 5% respectively at room temperature. A eutectic reaction takes place between 60%A and 40%B at 300C. a) Draw the phase diagram of A-B and label all the important points and fields. b) Find the temperature at which a 25%A-75%B alloy starts and completes solidification. c) Find the temperature at which the same alloy is composed of 50% liquid and 50% solid. d) Draw the cooling path of the alloy undergoing invariant reaction.	5	3	1,2,4,5	5	**1,2,3**	**5**
12.	Define 'green compact' in powder metallurgy and explain its properties.	2	4	1,2	2	**1,12,13**	**2**
13.	Write a part program for the component given below.	3	4	5,10	3	**3,13**	**3**

							
414.	Why it is not recommended to reuse abrasive particles in Abrasive Jet Machining?	**2**	4	1,2,6		**7,9**	**4**
15.	With a relevant neat sketch explain the laminated object manufacturing techniques. List its merits and demerits	**5**	4	1,2,3,5		**1,3,4,13**	**4**

MARK DISTRIBUTION (CO-wise & BTL-wise)

CO / CLO	Total Marks
1	***6***
2	***6***
3	***9***
4	***29***
5	-

BL	Total Marks
2	22
3	6
4	17
5	5